1. 局部变量和环境变量截图：

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var1="welcome to use local variable"
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var2="I am global variable"
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ export var2
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ bash
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var1"

yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var2"
I am global variable
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ exit
exit
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var1"
welcome to use local variable
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var2"
I am global variable
```

2. 引号的作用截图：

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo *
archlab-handout B24040413张岳晨lab2.docx B24040413张岳晨 lab3.docx B24040413张岳晨lab4.docx B24040413张岳晨lab5.docx bomb93 bomb93. bomb93. (1) bomb93
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "*"
*
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var3="*"
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo $var3
archlab-handout B24040413张岳晨lab2.docx B24040413张岳晨 lab3.docx B24040413张岳晨lab4.docx B24040413张岳晨lab5.docx bomb93 bomb93. bomb93. (1) bomb93
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var3"
*
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo '$echo'
$echo
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var4='ls -l'
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var4=`ls -l`
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "$var4"
total 2628
-rw------- 1 yoyo yoyo  475094 Jun  1 20:43 B24040413张岳晨 lab3.docx
-rw------- 1 yoyo yoyo 1527055 Apr 20  2026 B24040413张岳晨lab2.docx
-rw------- 1 yoyo yoyo  242104 Jun 12 21:32 B24040413张岳晨lab4.docx
-rw------- 1 yoyo yoyo  141801 Oct  9 10:49 B24040413张岳晨lab5.docx
drwxrwxr-x 2 yoyo yoyo    4096 Aug 22  2024 archlab-handout
drwxrwxr-x 2 yoyo yoyo    4096 Apr 20  2026 bomb93
drwx------ 3 yoyo yoyo    4096 Apr 18  2026 bomb93.
drwx------ 3 yoyo yoyo    4096 Apr 20  2026 bomb93. (1)
-rw-rw-r-- 1 yoyo yoyo   40960 Apr 15  2026 bomb93.tar
-rw------- 1 yoyo yoyo     719 Oct  9 10:17 copy.ys
drwxrwxr-x 2 yoyo yoyo    4096 May 30  2024 linklab
-rw-rw-r-- 1 yoyo yoyo    1704 Jun  9 10:04 linklab.tar
-rw-rw-r-- 1 yoyo yoyo   32304 May 22 21:33 printf.so
-rw-rw-r-- 1 root root     876 Apr  9  2026 sources.list
-rw------- 1 yoyo yoyo     568 Oct  9 10:17 sum.ys
-rw-rw-r-- 1 yoyo yoyo  184320 May 22 21:32 target310.tar
-rwxrwxr-x 1 yoyo yoyo       0 Oct 16 08:43 test1
-rw-rw-r-- 1 yoyo yoyo      69 Oct 16 08:43 test1.c
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ var5=ls
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ echo "`$var5`"
B24040413张岳晨 lab3.docx
B24040413张岳晨lab2.docx
B24040413张岳晨lab4.docx
B24040413张岳晨lab5.docx
archlab-handout
bomb93
bomb93.
bomb93. (1)
bomb93.tar
copy.ys
linklab
linklab.tar
printf.so
sources.list
sum.ys
target310.tar
test1
test1.c
```

3. shell 程序设计截图：

(1)

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ #!/bin/bash
hour=$(date +%H)

case $hour in
0[1-9]|1[01])
    echo "Good morning hour=$hour"
    ;;
1[2-7])
    echo "Good afternoon hour=$hour"
    ;;
*)
    echo "Good evening hour=$hour"
    ;;
esac
Good afternoon hour=15
```

(2)

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ #!/bin/sh
echo "Enter the first integer:"
read first
echo "Enter the second integer:"
read second
if [ "$first" -gt "$second" ]
 then
 echo "$first is greater than $second"
 elif [ "$first" -lt "$second" ]
then
 echo "$first is less than $second"
else
 echo "$first is equal to $second"
fi
Enter the first integer:
20
Enter the second integer:
22
20 is less than 22
```

(3)

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ #!/bin/bash
smallest=10000
for i in 8 2 18 0 -3 87
do
    if test $i -lt $smallest
    then
        smallest=$i
    fi
done
echo $smallest
-3
```

(4)

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ #!/bin/bash
count=0
for i in *
do
    if test -x "$i"
    then
        count=`expr $count + 1`
    fi
done
echo Total of $count files executable
Total of 6 files executable
```

(5)将脚本保存在文件中，赋予执行权限，后输入执行

```
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ nano prime.sh
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ chmod +x prime.sh
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ ./prime.sh 8
8 is not a prime!
yoyo@yoyo-VMware-Virtual-Platform:~/桌面$ ./prime.sh 13
13 is a prime!
```